DATA 205 Capstone Project Plan

Financial Markets and Investing | Youssef Mezni

Data

Nasdaq Historical Stock Data. Source: Nasdaq Historical Data. Separate files for AAPL, MSFT, GOOGL, and META contain 2,512 trading-day rows per company from September 2016 to September 2026. Each row is one trading day with Date, Close/Last, Volume, Open, High, and Low. Weekends and market holidays do not appear because they are not trading days.

SEC EDGAR Company Facts API. Source: SEC Company Facts API. The cleaned dataset has 156 company-quarter observations for the same four companies, covering approximately 2016-2026. Each row is one company-quarter with Revenue, Net Income, Operating Income, R&D, Assets, Liabilities, Cash, and reporting/filing information. The smaller row count is expected because financial statements are quarterly rather than daily.

Research Goal and Questions

Main goal: Investigate whether company financial information can help explain and possibly predict short-term stock performance, while comparing that with what can be learned from historical stock-market data alone.

1. Can historical stock prices help predict short-term stock performance?
2. Does prediction performance change across different future time horizons?
3. Does adding trading volume improve the result compared with historical prices alone?
4. Which financial variables are most strongly related to quarterly stock performance?
5. Do profitability variables show stronger relationships than balance-sheet variables?
6. Are the relationships different across AAPL, MSFT, GOOGL, and META?
7. Can financial information help distinguish whether quarterly stock performance goes Up or Down?
8. How does stock performance behave around major external events, and can those events explain unusual movements or model errors?

Tools, Methods, and Resources

I will work mainly in Python using Jupyter/Colab. The workflow includes API extraction, cleaning/wrangling, EDA, visualization, feature engineering, modeling, and evaluation. The final SEC-stage model is not locked in yet and will be selected after EDA.

Q1: Nasdaq - ARIMA using historical stock prices.
Q2: Nasdaq - compare ARIMA results across different prediction horizons.
Q3: Nasdaq - ARIMAX with Volume, compared with ARIMA.
Q4: SEC + Nasdaq - calculate quarterly stock returns and examine relationships with SEC variables.
Q5: SEC + Nasdaq - use scatter plots/correlations to compare groups of financial variables; choose the statistical model after EDA.
Q6: SEC + Nasdaq - compare the four companies separately.
Q7: SEC + Nasdaq - create an Up/Down outcome from quarterly return and use an appropriate classification model.
Q8: Nasdaq + event dates - use event/time-series visualizations around selected major events if time permits.

Risks and Mitigations

Weak or no relationship: Report it honestly; a weak relationship still answers the research question.
External events: Use context for major events such as COVID, market shocks, interest-rate changes, or major industry news when interpreting unusual periods.
Model overfitting: Evaluate models on data not used to fit them and compare out-of-sample performance.
Company-specific behavior: Compare the four companies separately instead of assuming one relationship applies to all.

Milestones
Oct. 31: Datasets locked; cleaning and wrangling complete.
Nov. 7: Layer 1 complete - Nasdaq analysis, ARIMA, and ARIMAX + Volume.
Nov. 17: Layer 2 complete - SEC financial-variable analysis and company comparisons.
Nov. 30: Layer 3 complete - quarterly return and Up/Down outcome created; final modeling/testing complete.
Dec. 5: Final analysis, evaluation, conclusions, and visualizations complete.
Dec. 8: Report, presentation, code/GitHub, citations, and final review complete.
Dec. 15: Final presentation.
Dependencies: Data cleaning must be completed before modeling. Layer 1 uses Nasdaq independently; Layer 2 depends on the cleaned SEC data; Layer 3 depends on linking SEC quarters with Nasdaq stock-return outcomes. Final conclusions depend on all three layers.